

Software Requirements Specification (SRS)

1. Introduction, Scope & User Classes

1. Introduction

1.1 Purpose

The purpose of this document is to define the requirements for a Multi-Platform (Mobile & Web) E-commerce Marketplace dedicated to used electronics. The application aims to solve the "Trust Deficit" in the current peer-to-peer market in Addis Ababa by introducing a **Managed Escrow System**, where the platform holds funds until the buyer verifies the device's condition.

1.2 Product Vision

To become the safest marketplace for buying and selling second-hand electronics (Smartphones, Laptops, Tablets) in Ethiopia, replacing informal Telegram groups and Facebook listings with a structured, verified, and secure application.

1.3 Scope of MVP (Minimum Viable Product)

- **Target Audience:** Residents of Addis Ababa and surrounding sub-cities.
- **Product Categories:** Strictly limited to Consumer Electronics (Mobiles, Tablets, Laptops/PCs). Future versions may include accessories.
- **Platform:**
 - **Mobile App:** Flutter (Android/iOS) for Buyers and Sellers.
 - **Backend:** Node.js (Express) & MongoDB.
 - **Admin Panel:** Web-based (React or Flutter Web) for internal management.
- **Languages:** English (Primary) and Amharic (Secondary).

2. General Description

2.1 The Core Problem

- Buyers fear purchasing "brick" (broken) devices or stolen goods.
- Sellers face "time-wasters" and lack a centralized platform to showcase inventory.
- Lack of standardized pricing makes it hard to know the true value of a used device.

2.2 The Solution: Escrow Logic

Unlike standard classifieds, this system intervenes in the payment:

1. Buyer pays the **App**.
2. Seller delivers the item.

3. Buyer inspects within a 24-hour window.
4. App releases funds to the Seller.

3. User Classes and Characteristics

The system defines distinct user roles with specific permission levels.

A. Guest (Unregistered User)

- **Description:** A user who has downloaded the app but has not signed in.
- **Capabilities:**
 - View "Home" and "Search" screens.
 - Filter products by price, brand, and condition.
 - Change App Language (English/Amharic).
- **Limitations:** Cannot view Seller contact info, cannot chat, cannot buy.

B. Standard User (The "Dual" Role)

- **Description:** The default verified account. Functions as both a Buyer and a Seller.
- **Verification Requirement:** Phone Number (OTP) + Email.
- **Capabilities:**
 - **Buying:** Add to cart, make secure payments, chat with sellers, rate transactions.
 - **Selling (Basic):** List items for sale (requires strict categorization).
 - **Wallet:** Manage an in-app balance (Credits from sales).
- **Security Level:** Low/Medium.

C. Verified/Premium Seller (Optional for Future, included in Design)

- **Description:** Intended for local shop owners or high-volume traders.
- **Verification Requirement:** Government ID (Kebele ID/Passport) + Business License (Trade License).
- **Capabilities:**
 - "Verified" Blue Badge on profile.
 - Priority listing in search results.
 - Higher withdrawal limits from the Wallet.

D. System Administrator

- **Description:** Internal staff managing the platform operations.
- **Capabilities:**
 - **User Management:** Ban users, Approve Verified Seller requests.
 - **Content Moderation:** Remove illegal or incorrect listings.
 - **Financial Control:**
 - Approve "Withdrawal" requests from Sellers.
 - Resolve Disputes (decide if money goes back to Buyer or to Seller).

4. Functional Requirements

4.1 Authentication & User Verification

The system must ensure a high level of trust by enforcing identity checks before transactions.

- **FR-01 (Sign Up/Login):**
 - The system shall allow users to register using Email and Password.
 - The system shall require Phone Number verification via OTP (One Time Password) upon registration.
- **FR-02 (Identity Verification - Level 2):**
 - *Constraint:* Users can browse freely, but to **Buy** or **Sell**, they must achieve "Verified Status."
 - The system shall provide a form for users to upload a clear image of a Government ID (Kebele ID, Passport, or Driver's License).
 - The system shall store the ID image securely (using Multer/Cloudinary) and set the user status to VERIFICATION_PENDING.
 - The system shall allow Admins to view pending IDs and approve/reject them.

4.2 Product Listing (Seller "Strict Wizard")

To maintain high data quality, the system prevents free-text titles.

- **FR-03 (Category Selection):** The seller must select a main category: *Mobile Phone*, *Tablet*, or *Laptop/Computer*.
- **FR-04 (Device Details):**
 - Based on the category, the system shall provide dropdown menus for **Brand** (e.g., Apple, Samsung) and **Model** (e.g., iPhone 13 Pro).
 - The system shall require technical specs: Storage (64GB - 1TB) and RAM (for PCs).
- **FR-05 (Condition Grading):** The seller must select one of the following descriptive statuses:
 1. **Brand New** (Sealed box).
 2. **Like New** (Open box, no scratches).
 3. **Good** (Minor scratches, fully functional).
 4. **Fair** (Visible wear/dents, fully functional).
 5. **For Parts** (Broken/Defective).
- **FR-06 (Images):** The system shall require a minimum of 3 images and a maximum of 10.
- **FR-07 (Pricing):** Input field for Price in ETB. The system may optionally suggest a price range based on historical data (Future feature).

4.3 Search & Discovery (Buyer Experience)

- **FR-08 (Advanced Filtering):** Users shall be able to filter results by:
 - Price Range (Min - Max).
 - Brand & Model.
 - Condition (e.g., "Exclude Broken items").
 - Location (Sub-city level).
- **FR-09 (Sorting):** Sort by Price (Low/High) and Newest Listed.

4.4 In-App Communication

- **FR-10 (Chat System):**
 - Buyers can initiate a chat from the Product Page.
 - The system shall support text and image sharing (for requesting more device photos).
 - **Safety Warning:** The chat interface shall display a persistent warning: *"Do not send money outside the app. You will lose your buyer protection."*

4.5 Order & Escrow Management (The Core Logic)

This module handles the money flow and trust mechanism.

- **FR-11 (Checkout):**
 - Buyer clicks "Buy Now" and selects a delivery method (Meetup or Shipping).
 - Buyer transfers funds. The system updates the Order Status to ESCROW_HOLD.
- **FR-12 (Seller Notification):**
 - The Seller receives a notification: *"Funds secured by App. Please deliver the item."*
 - The Seller cannot withdraw this money yet.
- **FR-13 (Delivery Confirmation):**
 - **Option A (Shipping):** Seller enters a tracking number/courier name. Status changes to SHIPPED.
 - **Option B (Meetup):** Seller scans the Buyer's QR Code (generated by the app) upon meeting. Status changes to DELIVERED.
- **FR-14 (Inspection Window):**
 - Once status is DELIVERED, a **24-Hour Timer** starts.
 - During this window, the Buyer can click "Report Issue" to pause the timer (Status: DISPUTE_OPEN).
- **FR-15 (Funds Release):**
 - If the Buyer clicks "Accept Item" OR the 24-hour timer expires without a dispute:
 - The system automatically moves funds from the "Escrow Account" to the "Seller's Wallet."
 - Order Status changes to COMPLETED.

4.6 Wallet & Financials

- **FR-16 (Wallet View):** Users can see their "Available Balance" (from sales) and "Pending Balance" (in Escrow).

- **FR-17 (Withdrawal):**
 - Users can request a withdrawal to a Bank Account or Telebirr.
 - This creates a "Withdrawal Request" ticket for the Admin.

5. Non-Functional Requirements

5.1 Performance & Response Time

- **NFR-01 (API Latency):** The system shall respond to user requests (e.g., search, loading product details) within **2 seconds** under normal network conditions (3G/4G).
- **NFR-02 (Image Optimization):** Product images uploaded by users must be automatically compressed by the server to ensure individual image sizes do not exceed **200KB** without significant loss of visual quality.
- **NFR-03 (Concurrent Users):** The backend system shall support at least **1,000 concurrent active users** without service degradation during the MVP phase.

5.2 Security & Data Privacy

- **NFR-04 (Data Encryption):**
 - All sensitive user data (passwords, PI) must be encrypted at rest in the database.
 - Passwords must be hashed using strong algorithms (e.g., bcrypt) before storage.
- **NFR-05 (Communication Security):** All data transmitted between the Flutter Client and the Node.js Server must be encrypted using **HTTPS/TLS 1.2+**.
- **NFR-06 (Authentication):**
 - API access must be secured using stateless tokens (e.g., JWT).
 - Sessions must auto-expire after a predefined period (e.g., 7 days) requiring re-login.
- **NFR-07 (Input Validation):** The system must validate all user inputs on the server side to prevent NoSQL Injection and Cross-Site Scripting (XSS) attacks.

5.3 Reliability & Availability

- **NFR-08 (Availability):** The system should aim for **99.0% uptime** during business hours (8:00 AM - 10:00 PM EAT).
- **NFR-09 (Transaction Integrity):** Financial transactions (Escrow logic) must be atomic. If a system failure occurs during a payment or fund release, the system must rollback to the previous consistent state to ensure no money is lost or duplicated.

5.4 Usability & Localization (Ethiopian Context)

- **NFR-10 (Language Support):** The mobile application interface must support dynamic switching between **English** and **Amharic** without restarting the app.

- **NFR-11 (Currency Display):** All monetary values must be displayed in **ETB (Ethiopian Birr)** formatted with commas (e.g., 50,000 ETB).
- **NFR-12 (Error Handling):** Error messages presented to the user must be descriptive and non-technical (e.g., "Network Error" instead of "Error 500").

5.5 Compatibility

- **NFR-13 (Mobile OS):** The Flutter application must function correctly on:
 - **Android:** Version 8.0 (Oreo) and above.
 - **iOS:** iOS 14 and above.
- **NFR-14 (Network Resilience):** The app must handle intermittent network connectivity gracefully (e.g., caching the last viewed products for offline viewing).